



RA2 – Computer Controlled Air Conditioning Unit

The RA2 is a fully instrumented bench mounted air conditioning duct for teaching all aspects of air conditioning systems.

The RA2 duct is a bench top unit fabricated from clear acrylic and mounted on a painted mild steel support frame.

PRODUCT CODE: RA2

Categories: RA Series - Refrigeration & Air Conditioning, RA2 - Computer Controlled Air Conditioning Unit

Tags: Academic Analysis , Air Flow Sensor , Armfield , Axial Fan , Chiller , Computer Controlled Air Conditioning Unit , Dehumidification , Dehumidifier , Duct , Enthalpy , Final Heating Stage , Heating Elements , Humidification , Humidity Ratio , Psychrometric , Ra2 , Refrigeration And Air Conditioning , Reheater , Relative Humidifier , Relative Humidity , Temperature , Thermodynamics , Vocational Training

DESCRIPTION

The RA2 is a fully instrumented bench mounted air conditioning duct for teaching all aspects of air conditioning systems.

The RA2 duct is a bench top unit fabricated from clear acrylic and mounted on a painted mild steel support frame.

An axial fan moves air through the duct. Heating elements are used to heat the air. Humidification is provided by steam from the boiler.

The steam boiler for the humidifier, the compressor and condenser for the refrigeration system and the electronic control box are all mounted beneath the duct.

Temperature and RH sensors record the temperature and relative humidity at every stage of the operation.

The air flow is measured by an electronic sensor. The mains supply voltage is monitored in the equipment to allow calculation of effective heater powers.

TECHNICAL SPECIFICATIONS

- **Transparent duct:** (200mm x 200mm) for process visibility
- **Pre-heater:** 400w
- **Re-heater:** 200w
- **Humidifier total volume:** 5ltr
- **Humidifier unit:** 2kW
- **Evaporator:** 500w @20°C
- **Air velocity sensor hot film anemometer:** 0-10m/s
- **Temperature / Relative humidity sensor x4: 10 to 100% relative humidity:**
 - Duct inlet
 - Before the evaporator
 - After the evaporator
 - Duct outlet

FEATURES & BENEFITS

- Transparent duct for visibility of the process and the components
- Pre-heaters, humidifier, chiller/dehumidifier and reheaters
- Temperature and relative humidity (RH) sensors at every stage of the process, plus an air flow sensor
- RH sensors calibrated for maximum accuracy
- Suitable for both vocational training and academic analysis of the thermodynamic principles involved
- Computer controlled with real time data logging of results (requires a PC not supplied by Armfield)
- Capability of using PID control for preheat, humidity and reheat allows stable conditions to be set up for investigations

Experimental content

- Gain an in-depth understanding of an air conditioning system by demonstrating the effects of essential air conditioning processes: cooling, heating, humidifying and dehumidifying
- Simulate different air conditioning environments and perform measurements to allow psychrometric data analysis
- To investigate and understand the use of psychrometric charts, understand relative humidity (RH) measurements and the effect of temperature on RH and understand the Humidity Ratio
- To investigate sensible heating of air in a duct
- To investigate the humidification of air
- To investigate the effect of vapour content and temperature on relative humidity
- To investigate dehumidification of air by cooling
- To investigate and understand the effect on enthalpy, understand relative humidity (RH) measurements and the effect of temperature on RH and understand the Humidity Ratio

Ordering Specification

- Air conditioning teaching system, complete with initial heating stage, humidifier, chiller/dehumidifier and final heating stage
- Transparent duct (200mm x 200 mm) for process visibility
- Computer controlled via USB interface
- Educational software including real-time data logging, graphics and data export
- Educational software, replicating the psychrometric chart calculations
- 4 sets of temperature and relative humidity measurements at the various stages of the process



- RH sensors come with calibration values which can be entered into the software for accuracy
- Electronic flowmeter to measure the air flow in the duct
- Dual control of boiler setting with a fast heat up setting and a gentle setting for control

Essential Accessories

The user must have a PC with a USB port, running Windows 7 or above.

Requirements

- Electricity supply: Single phase (see ordering codes)
- Distilled water required
- Software requires the user to have a PC running Windows 7 or above with a USB port.

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 1.2m³

Gross Weight: 150Kg

Dimensions

Length: 1.72m

Width: 0.40m

Height: 0.62m

Order Codes

- **RA2-A:** 220-240V/1ph/50Hz 13 Amps
- **RA2-B:** 110V/1ph/60Hz 25 Amps
- **RA2-G:** 220V/1ph/60Hz 13 Amps

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